MA342 Personal Project

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1 Introduction

Ludo is a four-player race game that has roots in the ancient Indian game *Pachisi*, which dates back to the sixth century CE. Players roll two dice to determine how far a piece can move, then choose which piece to advance. The probabilistic nature of movement makes optimal play analytically intractable. Knockouts add another layer of analytical complexity since, upon being knocked out, the piece returns to its home base and must be brought back on a future six, and lost progress can decide the game.

We adopt a Monte Carlo framework to compare the performance of two heuristic Ludo strategies. One strategy prioritizes hindering opponents via knockouts while the other aims to win by keeping pieces grouped together. We implement a simplified version of Ludo in MATLAB, define the strategies, and use repeated simulation to gauge their relative performance. We test the order and arrangement of strategies to determine their effect on win rate. We also assess game length and the number of knockouts each side inflicts to characterize their differences beyond win rate.

2 Model

2.1 Standard Rules and Simplifications

Ludo is played by four players, each with four pieces. The board consists of a 52-square loop and four home columns, each with six squares branching off the loop at the center of the board's side, one per player. All pieces start in their owner's base. On each turn, a player rolls two dice and moves one or two pieces forward by the rolled number of squares. A piece can only leave its base and enter the loop when a six is rolled on either die, after which it is placed at its owner's designated starting square. After going around the loop once, the piece enters its home column and moves towards the center. A player wins when all four of their pieces reach the center. If a piece lands on a square occupied by an opponent's piece, the opponent's piece is knocked back to base.

To simplify computation and to keep the strategy comparison interpretable, we adopt the following simplifications relative to the standard rule set:

1. **One die per turn.** Ludo is commonly played with two six-sided dice, meaning a player can move up to two pieces on a single turn. To avoid defining measures for allocating dice among pieces, we use a single die during the simulation.

2. **No extra turn on a six.** Players can roll again upon rolling a six. We drop this rule so each turn consists of exactly one roll and one move.

3. **Overshooting the finish counts as finishing.** Pieces are only allowed into the last square of the home column with an exact roll. We allow any roll that reaches or exceeds the final square to count as a finish.

4. **No safe squares.** Standard boards mark certain squares as safe from knockouts. We treat every loop square as "unsafe," meaning pieces can be knocked out anywhere except within their home columns.

5. **No blockades.** In some versions, two same-color pieces on a square form a blockade that others cannot pass. We do not implement blockades; same-color pieces share a square without obstruction, and contact with opposite-color pieces always results in a knockout.

These simplifications preserve the game's essence while removing rule clauses that would complicate strategy creation.

2.2 State and Notation

The game state is described by a 4×4 matrix S in which $S_{p,k}$ is the progress of player p's kth piece. A piece's progress can be any integer value between 0 and 57 with the following interpretation:

- $S_{p,k} = 0$: the piece is in base.
- $1 \leq S_{p,k} \leq 51$: the piece is on the main loop, with $S_{p,k} = 1$ being player p's own starting square.
- $52 \leq S_{p,k} \leq 56$: the piece is in player p's home column.
- $S_{p,k} = 57$: the piece is in the center and can no longer move.

The four players' starting squares are placed thirteen squares apart around the shared loop at the global positions $\{1, 14, 27, 40\}$. The global loop position of a piece with $1 \leq S_{p,k} \leq 51$ is

$$a_{p,k} = \big((s_p + S_{p,k} - 2) \bmod 52\big) + 1, \tag{1}$$

$s_p \in \{1, 14, 27, 40\}$ is the starting square of player p. Pieces in base or in a home column have no shared-loop position since they are safe. To indicate that a piece has been knocked out, its value is set to zero.

The game proceeds turn by turn in a fixed cyclic order $1 \to 2 \to 3 \to 4 \to 1 \to \cdots$. On each turn, a die is "rolled" by choosing a value d between 1 and 6 at random, and the active player moves their piece according to their rule set. The game ends when all the values in a row of S are 57.

2.3 Strategies as Decision Functions

A strategy is a function that takes the current game state, the active player's index p, and the die value, and returns the index k of the piece to move. If no legal move exists, like when all pieces are in base and $d \neq 6$, the strategy returns zero, and the turn moves to the next person. Both strategies in this study prefer to bring out a piece when they roll a six, but they will override this default when their strategy dictates.

3 Strategies

3.1 Aggressive

The first strategy, dubbed "Aggressive," is designed to hinder your opponents by knocking out as many pieces as possible. Its decision rule, in order of precedence, is as follows:

1. If $d = 6$, the player has a piece in base, and at least one of the player's on-board pieces could land a six-square move on an opponent, choose with probability $\frac{1}{2}$ between bringing one piece out and executing the knockout. If the knockout option is selected and multiple knockout-causing moves are available, choose uniformly at random among them.

2. Otherwise, if any valid move would knock an opponent's piece back to base (including a bring-out move that lands on an opponent sitting on the player's starting square), take that move, with random tie-breaking.

3. Otherwise, fall back to the baseline default.

Rule (1) captures the trade-off when a roll of six allows both a new piece to enter and a knockout to occur. Rule (2) ensures that knockouts are taken whenever available, even when no trade-off is involved.

3.2 Cohesive

The second strategy, named "Cohesive," is designed to win the race by keeping the player's pieces grouped together. We define the spread of a player's pieces as the difference between the maximum and minimum progress values among pieces currently on the board, excluding both base and finished pieces. The strategy's heuristic is as follows:

1. On a roll of six, if any piece is in base, bring it out (baseline default).

2. If at least three pieces are currently on the board, with probability 0.3 choose uniformly among the player's *intermediary* pieces — those that are neither the most-advanced nor the least-advanced.

3. Otherwise, evaluate every valid move and choose the one that minimizes the resulting spread, with ties broken at random.

The intermediary clause exists because strict spread minimization would always move the trailing piece. The 30% randomization keeps the strategy from collapsing to a single deterministic rule.

4 Experiments

4.1 Model Validation

We confirm the game mechanics work as intended by first running a homogeneous configuration in which all four players use the same strategy. As expected, each player's win rate converges to 25%, confirming that no slot enjoys an advantage in the absence of different strategies. Second, in two-versus-two configurations, the two strategies' per-slot win rates sum to 50%, since exactly one player wins each game and each strategy occupies two slots. Third, no game out of 10^5 simulated games hit the safety-net cap of 5000 turns, with the longest game being below 1000 turns, meaning all games will conclude at some point.

All experiments below use $N = 3000$ games per condition unless otherwise noted.

4.2 Baseline 2v2 Tournament

The headline experiment pits two Aggressive players against two Cohesive players in an alternating arrangement and reports the aggregate team win rate. Figure 1 shows the result.

The headline experiment uses an alternating strategy arrangement with a randomized starting player. Figure 1 shows the team win rates.

Aggressive's team wins the majority of games, and with knockouts unrestricted and starting squares unsafe, Cohesive's grouped pieces remain exposed to a player whose entire goal is to maximize knockouts.

4.3 Effect of Play Order

To test whether the baseline result is affected by play order, we ran four batches of games with strategies fixed, each batch using a different starting player. Figure 2 reports the result.

Panel (a) shows that a player's chance of winning is not improved by starting first; the win rate is independent of turn position. Panel (b) averages over the four starting-player conditions and finds all four positions within sampling noise of the 25% baseline.

The absence of an order effect seems counterintuitive initially — one expects the first player to have taken more turns by the end of the game. This can be explained by the cyclic turn rotation, which causes turns to be distributed nearly equally by the end of the game, so any advantage gained from having an additional roll becomes negligible by the end of the game.

4.4 Effect of Strategy Placement

The second potential confounder is the arrangement of strategies around the board. The starting squares are separated by thirteen squares on the loop, so two players sitting adjacent (e.g., slots 1 and 2) are closer to each other's pieces than two players sitting opposite

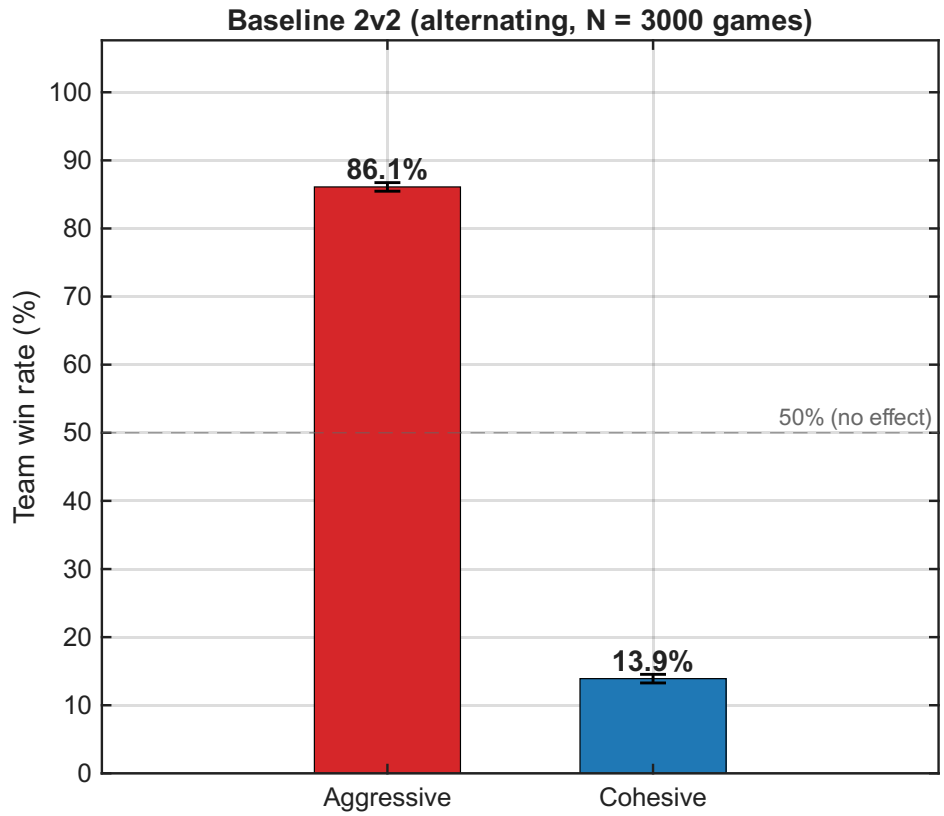


Figure 1: Team win rates from the baseline two-versus-two tournament with the alternating arrangement (Aggressive at slots 1, 3; Cohesive at slots 2, 4) and randomized starting player. The dashed line at 50% marks the chance baseline in the absence of a strategy effect. Each strategy occupies two of the four slots, so team win rates sum to 100% rather than 50%.

(slots 1 and 3), and because knockouts depend on relative loop position, the placement of strategies could affect the win-rate gap. To test this, we ran the six distinct two-versus-two arrangements of Aggressive and Cohesive.

Figure 3 displays the per-slot win rates, and Table 1 summarizes the same data alongside the corresponding game-length and knockout statistics used in the following sections.

Aggressive's high win rate over Cohesive is, again, independent of placement; no arrangement reverses the ordering. The alternating configurations `AG CO AG CO` and its mirror yield the largest Aggressive advantage, while bracketing arrangements (`AG CO CO AG` and `CO AG AG CO`) reduce the gap marginally, likely due to the fact that the strategies that do not seek knockouts are closest together, leading more of Cohesive's pieces to be safe from intentional knockouts. Because none of the arrangements significantly alter the results, the strategy arrangement has a minimal effect on the overall win rate.

4.5 Game Length

We sought to ascertain whether the winning strategy also produces shorter games. Figure 4 combines the game-length data of all six placements and divides it by the winning team,

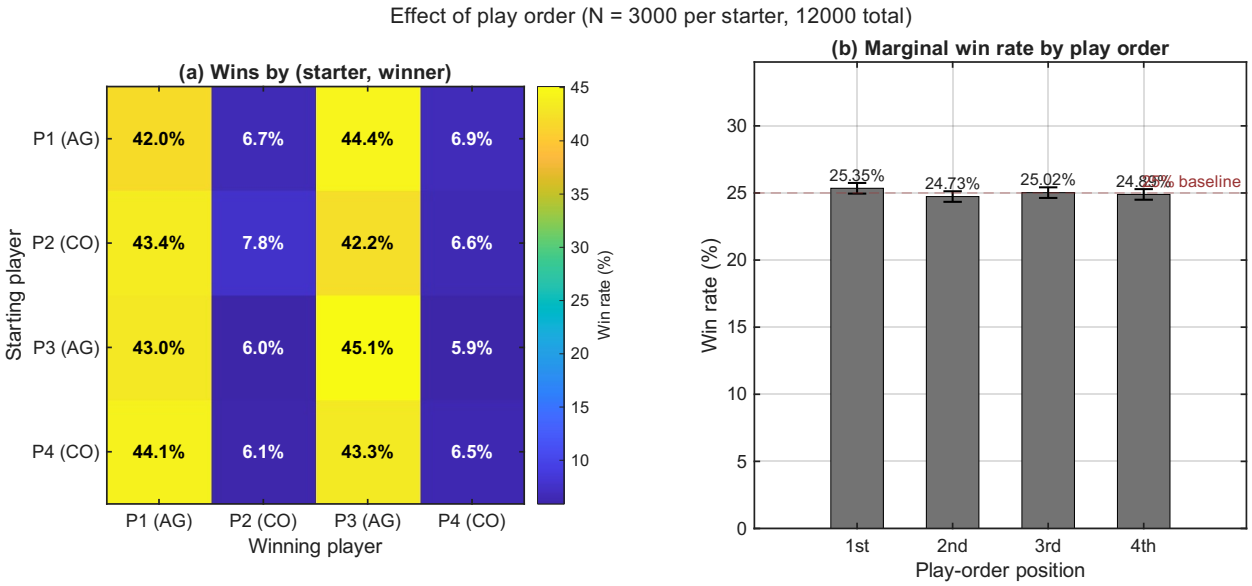


Figure 2: Effect of play order with strategies fixed at slots. (a) Wins broken down by starting player and winning player. (b) Win rate by play-order position averaged over the starting-player condition so that each strategy spends equal time at each position. Under no order effect, all four bars sit near 25%.

producing a normalized distribution of games that Aggressive won and another for games that Cohesive won.

The two distributions overlap substantially, but their medians differ slightly. The games that Aggressive wins are slightly shorter on average than games that Cohesive wins. Two effects contribute to this:

1. When Aggressive is winning, its successful knockouts reset Cohesive's progress and broaden the gap, so the game ends earlier.

2. The only case when Cohesive manages to win despite Aggressive's interference is when the game has gone on long enough for Cohesive to have overcome multiple setbacks. This results in the Cohesive-winning distribution to be more right-skewed than the right tail of the Aggressive-winning distribution.

The asymmetry is modest in magnitude but consistent in direction.

4.6 Knockouts

Finally, we quantify the key difference that underlies the overall result: knockouts. Aggressive is defined by its pursuit of knockouts, whereas Cohesive's knockouts are incidental. Figure 5 reports the average number of knockouts each team's two players inflict in a single game across the six placements.

Aggressive's team inflicts roughly twice as many knockouts as Cohesive's team, and the gap is consistent across all six combinations. We see a similar pattern here as we did with the strategy placement: the alternating configurations produce slightly higher knockout counts

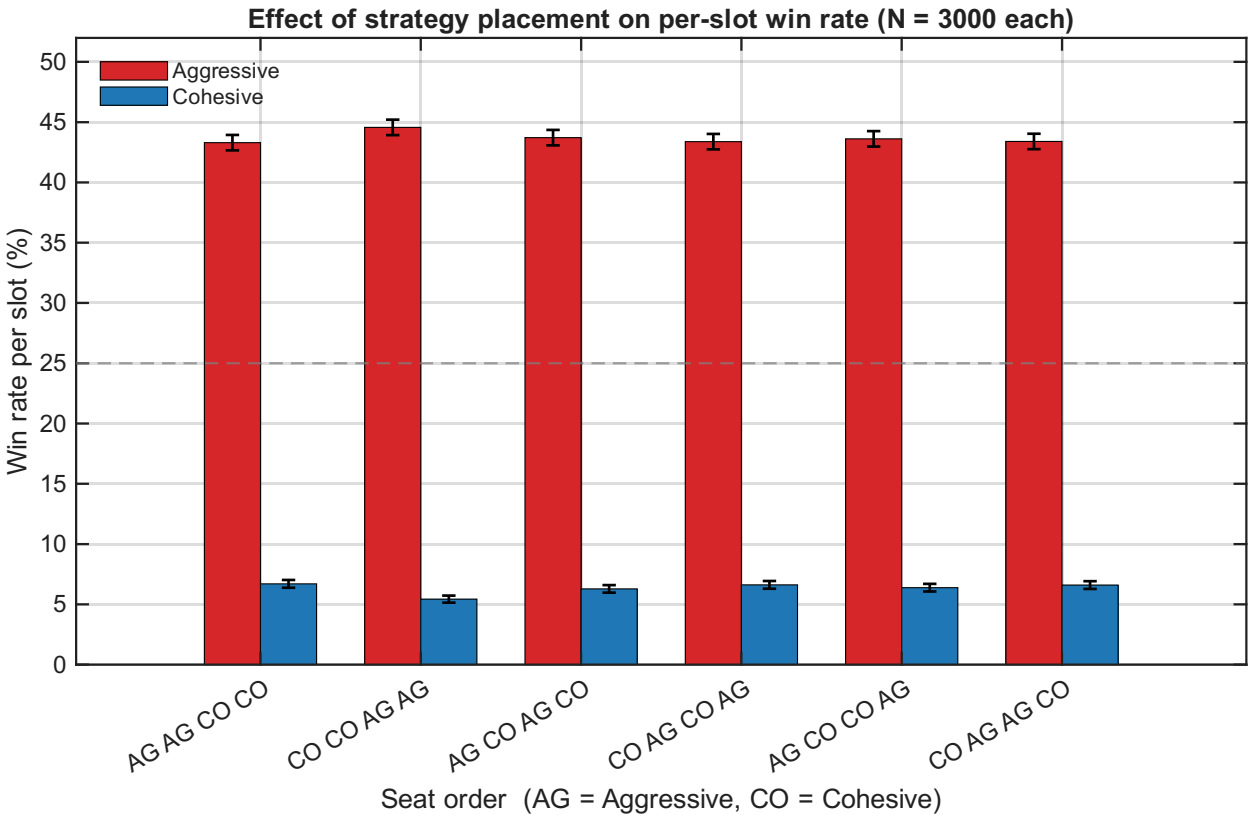


Figure 3: Per-slot win rate of Aggressive and Cohesive across the six two-versus-two placement arrangements. The starting player is randomized per game so that first player effects average out. Each pair of bars sums to 50%, since each strategy occupies two slots.

than bracketing or clustering (`AG AG CO CO`, `CO CO AG AG`) arrangements. This is consistent with the intuition that closer opponents are easier to knock out, but, as in the placement analysis, the variation is minimal.

5 Conclusion

We have implemented a Monte Carlo simulation of a simplified Ludo game and used it to compare two strategies: one that pursues knockouts whenever available and another that seeks victory by moving pieces as a herd. Across roughly 25,000 simulated games, the Aggressive team won a clear majority of baseline games, beat Cohesive across every placement tested, and inflicted roughly 10 times as many knockouts per game as Cohesive.

Two controls supported these conclusions. The play-order analysis ruled out a first-player bias, and the placement analysis showed only mild effects on the win-rate gap. The games Aggressive wins are also significantly shorter than games Cohesive wins, consistent with each strategy's design. The fact that Aggressive dominated games is likely inflated by the game's simplification. The omission of safe spaces allowed for knockouts to occur where they should not, and the lack of blockades left pieces that would normally be shielded vulnerable.

The most direct extension of this project would be to implement Ludo without omitting

Table 1: Per-placement summary statistics. "Per-slot win" is the average win rate of one of the two slots occupied by each team; values within a row sum to 50%. "KO/game" is the average number of knockouts each team's two players together inflict in a single game. Values shown are illustrative of a representative $N = 3000$ run.

Configuration	AG win (%)	CO win (%)	AG KO / game	CO KO / game
AG AG CO CO	43.3	6.7	43.50	24.85
CO CO AG AG	44.6	5.4	43.58	24.45
AG CO AG CO	43.7	6.3	44.76	23.81
CO AG CO AG	43.4	6.6	45.19	24.31
AG CO CO AG	43.6	6.4	43.04	24.50
CO AG AG CO	43.4	6.6	43.34	24.58

any rules, thereby eliciting the need to alter strategies to accommodate additional scenarios not present in the simplified version. Another extension would be to introduce strategies specifically designed to counter the already devised strategies. For instance, a strategy to counter Aggressive would be a defensive strategy that avoids landing within knockout distance of opponents. This would permit testing whether knockout avoidance can break the dominance pattern.

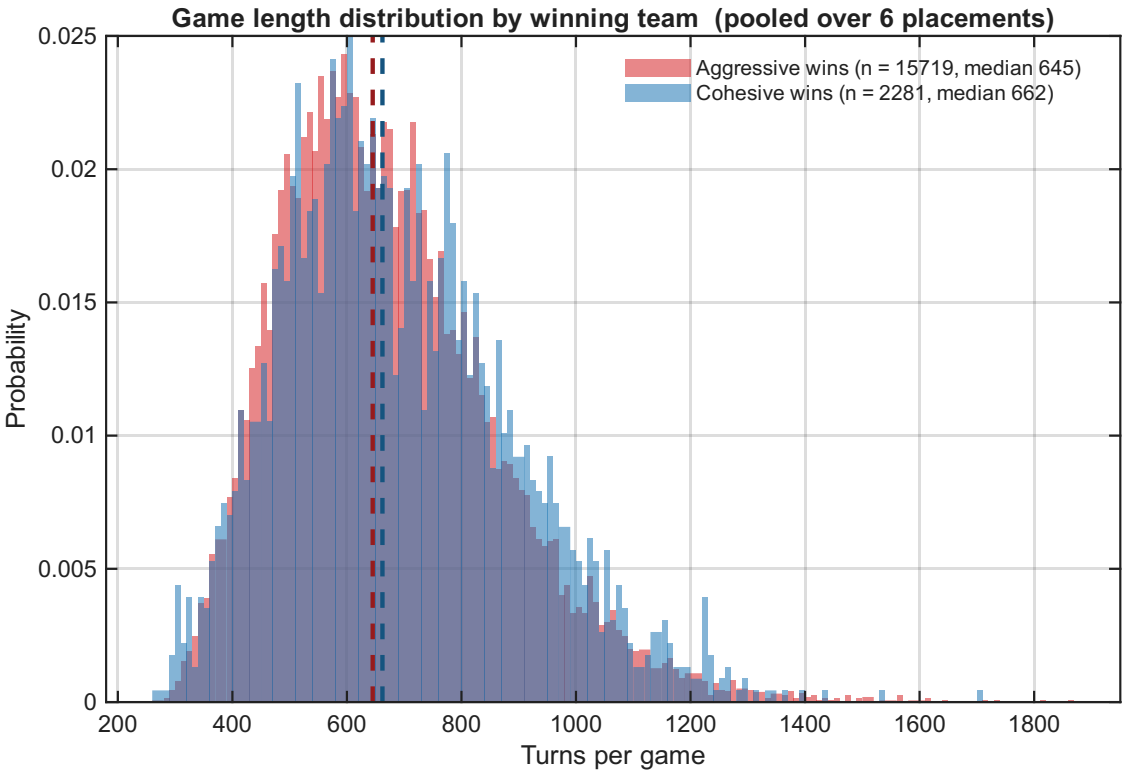


Figure 4: Distribution of game length pooled across the six placement configurations. The counts are normalized to probabilities so that the two distributions can be compared despite different sample sizes. Dashed vertical lines mark the medians.

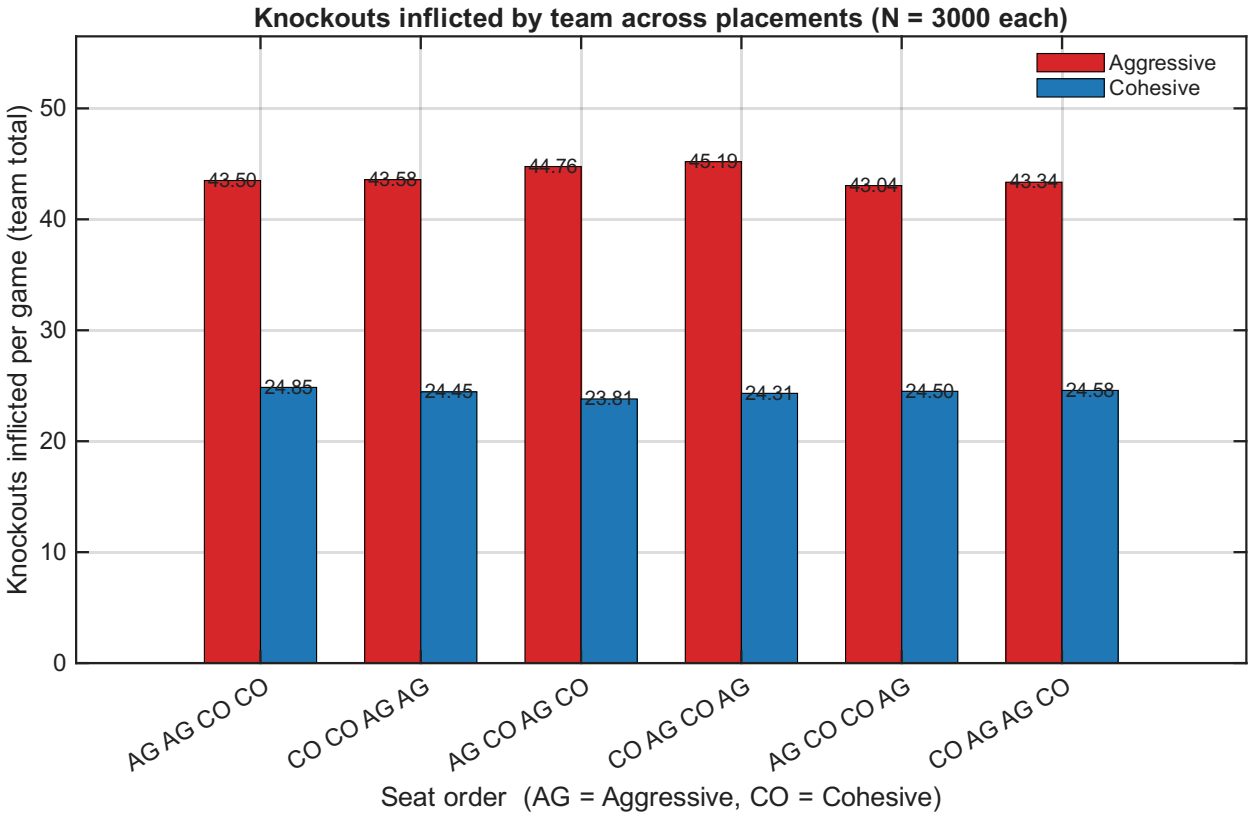


Figure 5: Average number of knockouts inflicted by each team's two players (combined) in a single game, across the six placement configurations. Aggressive is by design a knockout-seeking strategy; Cohesive's small but nonzero count reflects incidental knockouts that arise when a spread-minimizing move happens to land on an opponent.